

Photosynthesis in Higher Plants - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



✓ 6 PM :- Breathin

Good afternoon

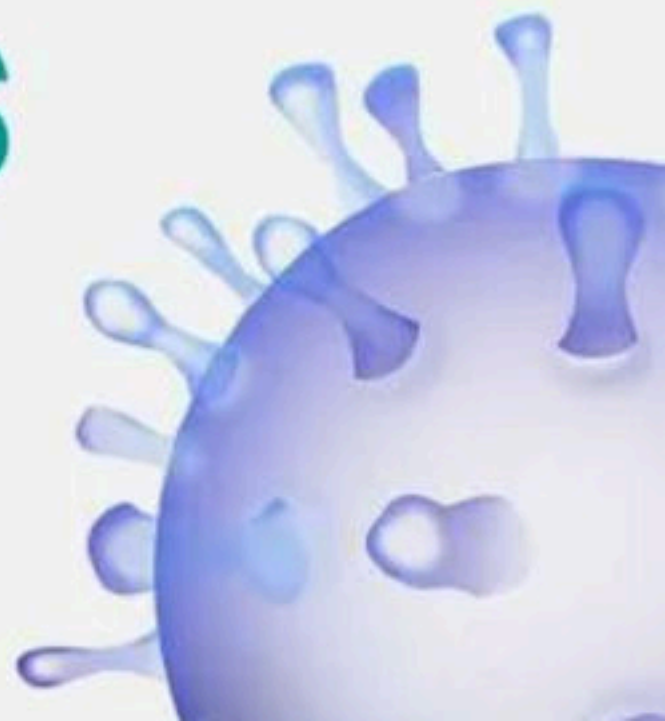
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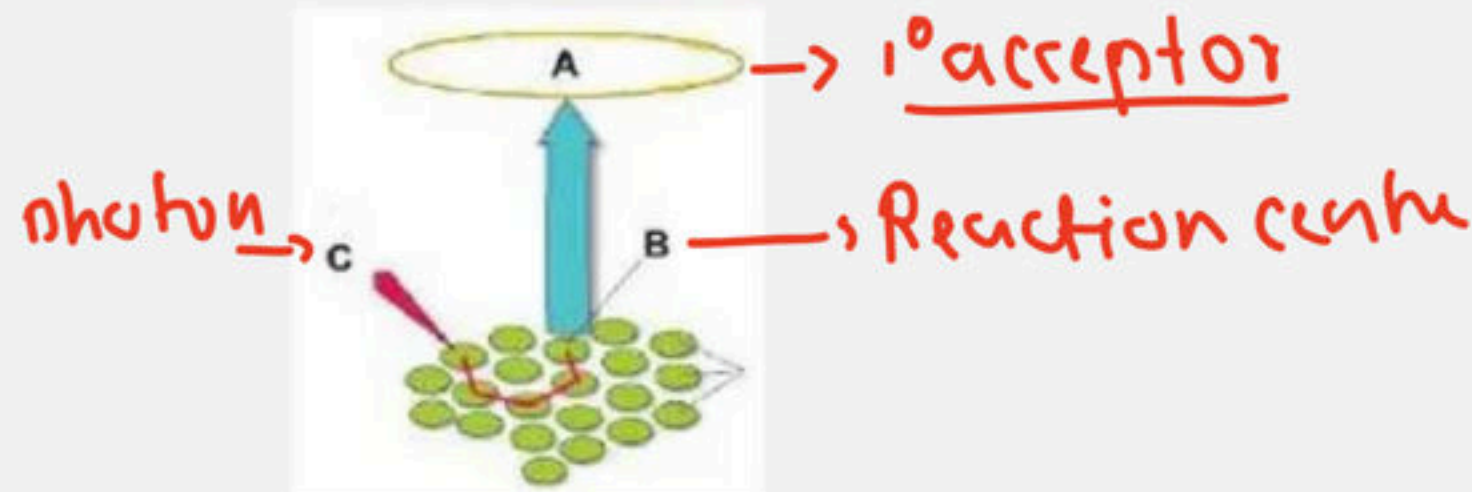
Photosynthesis in

Higher Plants



BIOLOGYLIVE

1. Choose the correct option according to given figure:-



(1) A – Primary acceptor , B – Reaction center, C – Pigment molecules.

✓(2) A –Primary acceptor, B– Reaction center, C – Photon

(3) A – Photon, B – Pigment molecules, C – Reaction centre

(4) A – Reaction center, B – Primary acceptor, C – Pigment molecules



2. Photosynthesis is ["]more in["] :

(1) Orange light

(2) Green light

✓ (3) Blue light *and Red light*

(4) Yellow light



3. As compared to a C₃ plant how many additional molecules of ATP are needed for net production of one molecule of hexose sugar by maize :

- (1) Two
- (2) Six
- (3) Zero
- (4) Twelve

$C_4 \text{ plant} = 30 \text{ ATP} + 12 \text{ NADPH}$
 $C_3 \text{ plant} = 18 \text{ ATP} + 12 \text{ NADPH}$
↓
C₄ plant
↓
Glucose



4. During photosynthesis electrons are continuously lost from the reaction centre of PSII. What source is used to replace these electrons:

(1) Sunlight

(2) O_2

✓ (3) H_2O

(4) O_3



5. The electron flow from PSII to NADP via ETS and PSI follows V.I.M.P

✓ (1) Uphill → Downhill → Uphill → Downhill

(2) Downhill → Uphill → Downhill → Uphill

(3) Downhill → Uphill → Uphill → Downhill

(4) Uphill → Downhill → Downhill → Uphill



6. Which statement is false :

(1) H_2S , not H_2O is involved in photosynthesis of purple sulphur bacteria ✓

(2) Light reactions are stopped in the absence of light ✓

✓ (3) Calvin cycle occurs in the ^{stroma} ~~grana~~ of chloroplast ✗

(4) ATP is produced during light reaction via chemiosmosis



7. Stromal lamellae lack NEET 2025

(1) PSII ✓

(2) PSI

(3) NADP reductase ✓

✓ (4) Both 1 and 3



8. A process that makes important difference between C₃ and C₄ plants is:

(1) Transpiration

(2) Glycolysis

(3) Photosynthesis

(4) Photorespiration

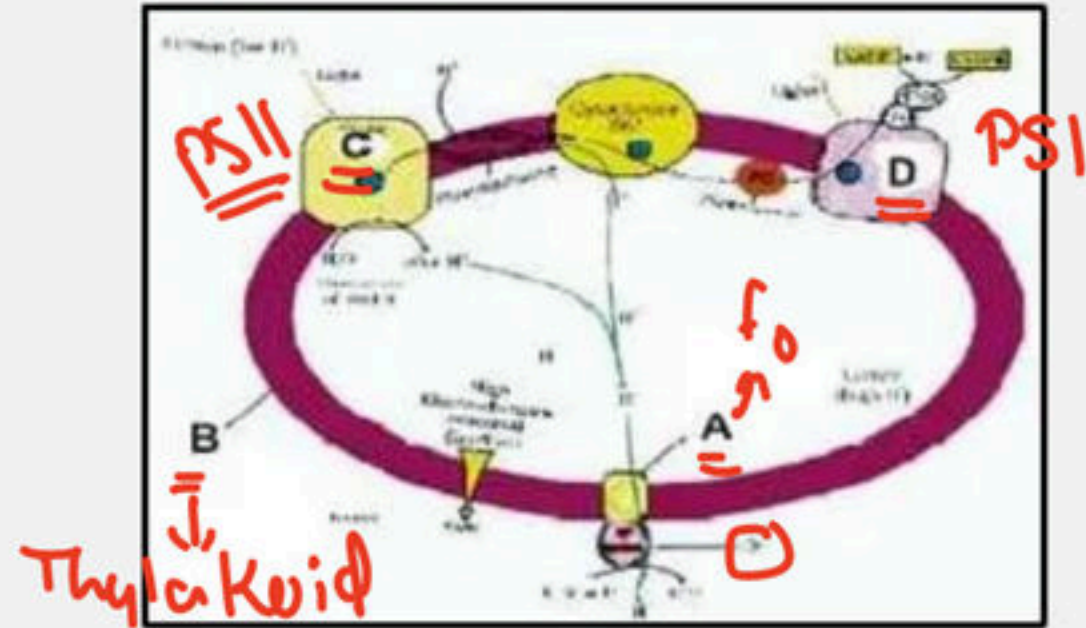
→ lack → C₄ plant

✓ ~~~~~



9. Study the pathway ATP synthesis through chemiosmosis given below:

In which of the following options correct for all the four blanks A, B, C and D are indicated.



- (1) A–F₁, B–Thylakoid membrane, C–Photosystem (I), D–Photosystem (II)
- (2) A–F₀, B–Thylakoid membrane, C–Photosystem (I), D–Photosystem (II)
- (3) A–F₁, B–Thylakoid membrane, C–Photosystem (II), D–Photosystem (I)
- ✓ (4) A–F₀, B–Thylakoid membrane, C–Photosystem (II), D–Photosystem (I)



10. Which statement is incorrect : NIFFI 2025

diff.

- ✓ (1) The C_3 and C_4 plants respond similarly to CO_2 concentration
- (2) At low light conditions, neither C_3 nor C_4 plants respond to high CO_2 concentration
- (3) C_4 plants attain saturation at much lower CO_2 concentration (about 360 $\mu l-l^{-1}$) than C_3 plants (about 450 $\mu l-l^{-1}$)
- (4) Current availability of CO_2 levels is limiting to the C_3 plants



11. Which statement are true for photosynthesis :

- (1) CO_2 reduction occur in dark reaction ✓
- (2) H_2O oxidation occur in light reaction ✓
- (3) Dark reaction is an enzymatic phase of photosynthesis ✓
- (4) All of these ✓



12. Kranz anatomy is related with :

- ✓ (1) C_4 pathway ↻
- (2) C_3 cycle
- (3) C_2 cycle
- (4) CAM cycle



13. Which of the following factors not affects photosynthesis

(1) number, size, age and orientation of leaves ✓

Imp

(2) mesophyll cells and ~~chromoplast~~ ^{chloroplast} ✗

(3) internal CO₂ concentration and amount of chlorophyll ✗

(4) 2 and 3

✓



14. Splitting of water in photosynthesis is called :

(1) Dark reaction

✓ (2) Photolysis = PSII

(3) Electron transfer

(4) Phototropism



15. Which of the following are produced by reactions that take place in the thylakoids and consumed by reactions in the stroma :

(1) CO_2 and H_2O

(2) NADP^+ and ADP

✓ (3) ATP and NADPH

(4) Glucose and O_2



16. The first electron carrier molecule in the chain from P700 to NADP⁺ is believed to be an :

(1) Iron protein

(2) Iron magnesium protein

✓ (3) Iron sulphur protein Fe-S complex

(4) Iron copper protein



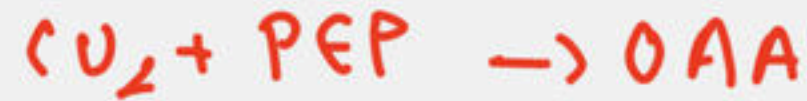
17. In the initial fixation of CO₂ in C₄ plants :

(1) CO₂ reacts with PGA

~~(2)~~ CO₂ reacts with PEP 

(3) CO₂ reacts with RuBP

(4) CO₂ reacts with RuMP



18. In pigment system II electrons from excited P680 accepted by :

- (1) Phaeophytin ↪
- (2) Plastocyanin
- (3) Plastoquinone ✗
- (4) AFeS



19. Cyclic photophosphorylation produces :

(1) NADPH

(2) ATP and NADPH → Non-cyclic

(3) ATP, NADPH and O_2

(4) ATP only



20. When O₂ binds to RuBisCO the products in Calvin cycle is :-

- (1) 2 Phosphoglycerate molecules
- (2) 2 phosphoglycolate molecules
- ☒ (3) One phosphoglycerate and one phosphoglycolate molecule ✓
- (4) 1 Phosphoglyceraldehyde molecules



11.11.2020

21. Which pigment system immediately donates e^- for the reduction of NADP :-

(1) PS II ✗

✓ (2) PS I

(3) CO_2

(4) Plastoquinone



22. In addition to the 12 molecules of NADP.H₂ the energy required for the synthesis of one mole of hexose by C₃ and C₄ pathway is :

(1) 18 molecules of ATP

(2) 30 molecules of ATP

✓ (3) 18 & 30 molecules of ATP respectively

(4) 30 & 18 molecules of ATP respectively

↓ ↓
18 ATP 30 ATP



23. The remarkable feature of photosynthesis is :-

- ✓ (1) Light energy is converted into chemical energy
- (2) Formation of sugars from formaldehyde
- (3) Formation of chlorophyll from inorganic salts
- (4) Oxidation of chlorophyll



24. ³ATP : ²NADPH₂ : ¹CO₂ consumption ratio during the photosynthesis in the C₃ plant :

✓ (1) 3 : 2 : 1

(2) 1 : 2 : 3

(3) 2 : 3 : 1

(4) 1 : 2 : 4



25. The enzyme that not found in C₃ cycle :

✓ (1) PEP carboxylase → cy

(2) RuBp carboxylase

(3) NADP reductase

(4) ATP synthase



26. Reaction centre of the photo system of photosynthesis forms by :

- (1) Single molecule of chlorophyll b
- ✓ (2) Single molecule of chlorophyll a
- (3) Chlorophyll b and Xanthophyll
- (4) Xanthophyll and carotenoid



27. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In C₄ plants photorespiration does not occur.

Reason (R) : C₄ plants have a mechanism that increase the concentration of CO₂ at the enzyme site.

choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

☒ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



28. At high light intensities : V.V. Imp

- (1) C_3 plants show increase in photosynthesis and C_4 show no effect
- (2) C_4 plants show increase in photosynthesis and C_3 show no effect
- ✓ (3) In both C_3 and C_4 plants shows increase rate of photosynthesis
- (4) Photosynthesis is not dependant on the intensities of light.



29. Which statement (s) is/are true : V. Imp

- ✓ (1) Dark reaction is more temperature sensitive than light reaction
- (2) Light reaction is more temperature sensitive than dark reaction
- (3) Light and dark both reactions are not sensitive to temperature
- (4) Dark reaction is sensitive to temperature but light reaction is not sensitive to light.

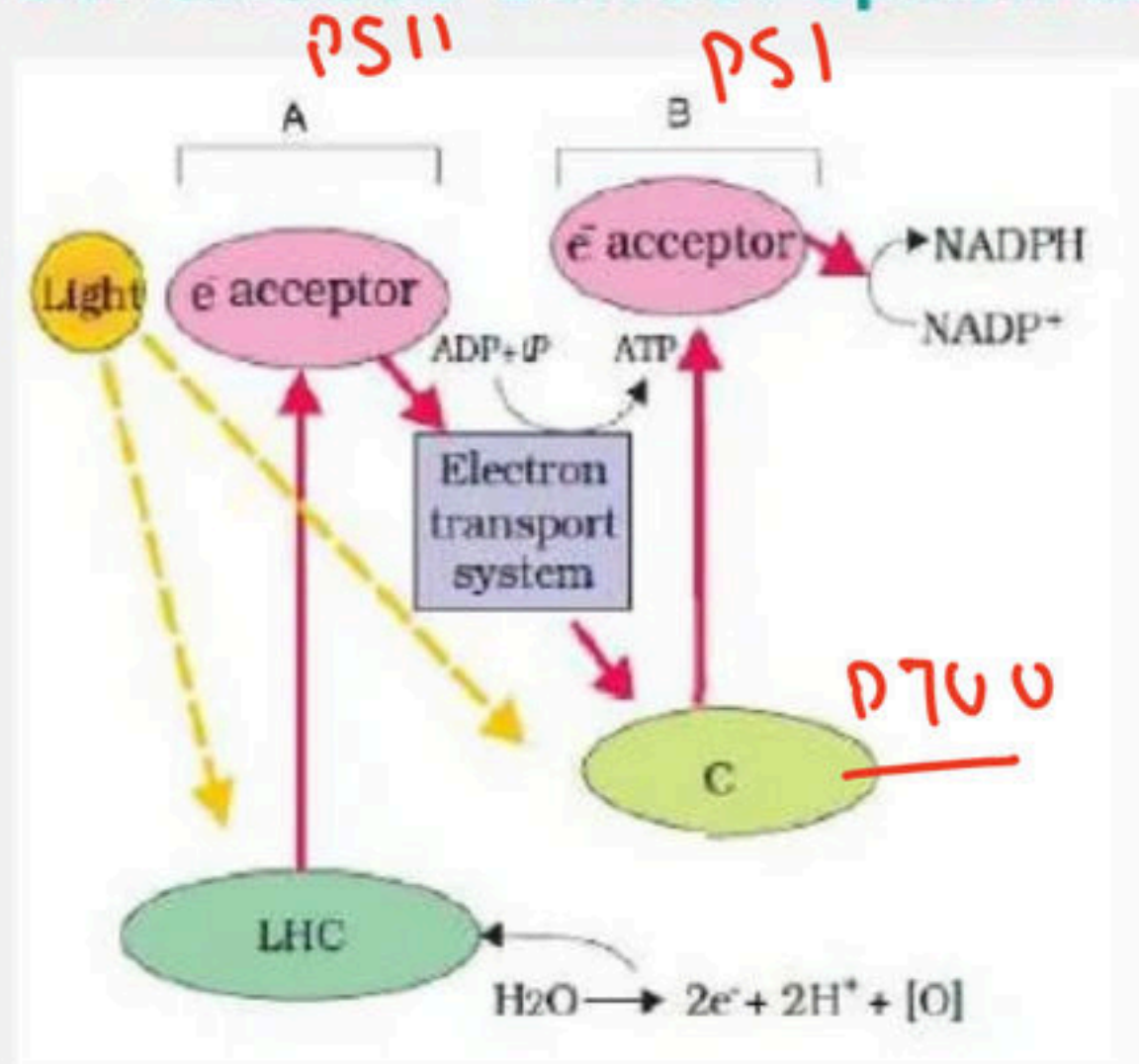


30. Which one of the following would not be a normally a limiting factor of photosynthesis :

- ✓ (1) Oxygen ✗
- (2) Light
- (3) Carbondioxide
- (4) Chlorophyll



31. Choose correct option according to given figure :



- (1) A- PS I. B- PS II, C- P 680
- ✓ (2) A- PS II. B- PS I, C- P 700
- (3) A- PS II. B- PS I, C- P 680
- (4) A- PS I. B- PS II, C- P 700



32. In photosynthesis, the light-independent reactions take place at :

(1) Photosystem II

✓ (2) Stroma matrix

(3) Thylakoid lumen

(4) Photosystem I.

↓
Dark



33. Which of the following is not affect the rate of photosynthesis :

(1) Light quality

✓ (2) Duration of exposure of light

(3) Light intensity

(4) None of these



34. Which of the following is incorrect :

- (1) In maize, calvin cycle occur in mesophyll cells ^{BSC}
- (2) Hatch and slack pathway occurs in sorghum ✓
- (3) During hatch and slack pathway aspartic acid formed in mesophyll cells
- (4) During calvin cycle, 3 – phosphoglycerate is formed in mesophyll cell



35. Which of the following the electron carrier is present on inner side of thylakoid membrane : NEA 2025

(1) Plastoquinone

✓ (2) Plastocyanin

(3) ATP synthase

(4) All of these



36. During the light reaction in photosynthesis, the following are formed:

(1) ATP and sugar

(2) Hydrogen, O_2 and sugar

✓ (3) ATP, hydrogen donor, and O_2

(4) ATP, hydrogen, and O_2 donor



37. What occurs during the light reaction of photosynthesis?

(1) Chlorophyll is produced ✗

✓ (2) Water splits to form 2H^+ and O_2

(3) CO_2 is given off as a waste

(4) Sugar is formed from CO_2 and water



38. Energy required for ATP synthesis in PS II comes from:

- ✓ (1) Proton gradient
- (2) Electron gradient ✗
- (3) Reduction of glucose
- (4) Oxidation of glucose



39. When CO₂ is added to PEP, the first stable product synthesized is:

- (1) Pyruvate
- (2) Glyceraldehyde-3-phosphate
- (3) Phosphoglycerate
- (4) Oxaloacetate 4C



40. Plants living in a hot, humid environment are:

(1) C_3 plants

(2) CAM plants

☒ (3) C_4 plants

(4) All of the above ✗



41. Which of the following would not be a limiting factor in photosynthesis?

✓ (1) O₂

(2) CO₂

(3) Chlorophyll

(4) Light



C₄ plant

42. In sugarcane, ¹⁴CO₂ is fixed in malic acid. Which enzyme fixes CO₂?

- (1) Ribulose phosphate kinase
- (2) Fructose phosphatase
- (3) Ribulose biphosphate carboxylase
- ☒ (4) Phosphoenol pyruvic acid carboxylase



43. Carbon dioxide is necessary for photosynthesis. The chemical used to remove this gas most effectively from entering a control apparatus is:

- (1) Distilled water
- (2) Sodium carbonate
- (3) Calcium oxide
- ✓ (4) Potassium hydroxide solution



44. Match the columns and find out the correct combination:

Substrate	Enzyme	Product
A. RuBP	<u>RuBisCO</u> x	C4 acid
B. <u>3-PGA</u>	<u>RuBisCO</u> →	<u>C3 acid</u> ✓
C. Phosphoenol pyruvate	<u>PEP kinase</u> x	C3 acid
D. <u>Oxaloacetate</u>	<u>Malate dehydrogenase</u> →	<u>C4 acid</u> ✓

Options:

(1) A and B (2) B and D ✓

(3) B and C (4) C and D



45 Match the columns and find out the correct combination.

Column I	Column II
A. Oxygen evolving	1. ATP synthesis complex
B. Proton gradient	2. <u>High oxygen concentration</u>
C. Photorespiration	3. Kranz anatomy
D. C4 plants	4. Photolysis of water

Options:

(a) A-2, B-4, C-3, D-1

(b) A-4, B-1, C-3, D-2

☒ (c) A-4, B-1, C-2, D-3

(d) A-2, B-1, C-4, D-3

28th
8^{PM}

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only

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46. Match the columns and find out the correct combination:

Column I	Column II
A. RuBisCO	1. Accessory pigment
B. PEPCase	2. Absorption maxima at 700 nm
C. PS II	3. C ₃ cycle
D. Carotenoids	4. Absorption maxima at 680 nm
E. PS I	5. C ₄ cycle

(a) A-2, B-4, C-1, D-3, E-5

(b) A-5, B-1, C-4, D-2, E-3

(c) A-3, B-1, C-2, D-5, E-4

(d) A-3, B-5, C-4, D-1, E-2



47. Incorrect statement is:

- (a) Except for plants in shade or in dense forests, light is a limiting factor for photosynthesis. Rarely
- (b) Internal factors of photosynthesis and respiration are dependent on the genetic predisposition and the growth of the plant.
- (c) At any point, the rate of the process is determined by the factor available at sub-optimum levels.
- (d) The C_3 plants show saturation at about 360 ppm CO_2 , while C_4 responds to increased CO_2 concentration and saturation is seen only beyond 450 ppm. sub-optimum

X

 C_3 plant

48. Incorrect statement is:

- (a) Water-splitting complex is physically located on the inner side of the membrane of the thylakoid. ✓ Wrong
- (b) The electrons needed to replace those removed from photosystem I are provided by PS-II. ✓
- (c) The splitting of water is associated with PS-II. ✓
- (d) Functioning of PS I is not necessary to synthesize NADPH_2 . ✗



49. Which of the following statements with respect to photosynthesis is/are correct?

A. Three carbons are present in the primary CO₂ acceptor in C₄ plants.

B. Temperature optimum for C₄ plant is 10-15°C. ✗

C. RUBP is the primary CO₂ acceptor in ~~C₄~~ plants. ✗
C₃

(a) A & C (b) A, B & C

☒ (c) A only (d) C only



50. Select the false statement. :-

- (a) ATP or NADH is not formed during photorespiration.
- (b) C₄-pathway of photosynthetic CO₂ fixation was discovered by Hatch and Slack.
- ☒ (c) Photorespiration results in maximum production of ATP. ✗
 → wasteful
- (d) In C₄ plants, phosphoenol pyruvate (PEP) acts as CO₂ acceptor.



51. During chemiosmotic coupling hypothesis oxidative phosphorylation proposes occurs for formation of adenosine triphosphate (ATP)

(1) There is a change in the permeability of the inner mitochondrial membrane toward adenosine diphosphate (ADP)

(2) High energy bonds are formed in mitochondrial proteins

(3) ADP is pumped out of the matrix into the intermembrane space

✓ (4) A proton gradient forms across the inner membrane



52. Match the following :

- | | |
|----------------|-------------------------------------|
| a) P700 | i) PS II |
| b) P680 | ii) PS I |
| c) P680 & P700 | iii) Cyclic photophosphorylation |
| | iv) Non-cyclic photophosphorylation |
- 

(1) a-ii, b-i, c-iii & iv

(2) a-ii, b-i & iii, c-iv

✓ (3) a-ii & iii, b-i, c-iv

(4) a-ii, b-iii, c-i & iv



53. Given below are two statements

Statement I : Lamellae of grana have both PS I and PS II ✓

PS I - 700 nm

✗ Statement II : Cyclic photophosphorylation occur when light of wave length above 680 nm are available for excitation. ✗

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- ✓ (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct



54. Given below are two statements

Statement I : Water splitting complex associated with PS II.

Statement II : H₂S is the Hydrogen doner for purple and green sulphur bacteria.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct ✓



55. Who used prism, white light, green alga, Cladophora and aerobic bacteria and plotted the action spectra for photosynthesis:

(1) Sachs

(2) Arnon

(3) Arnold

☒ (4) Englemann

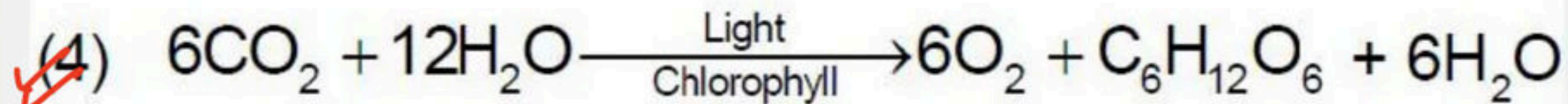
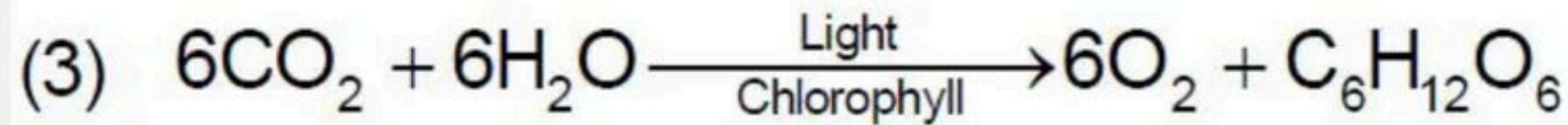
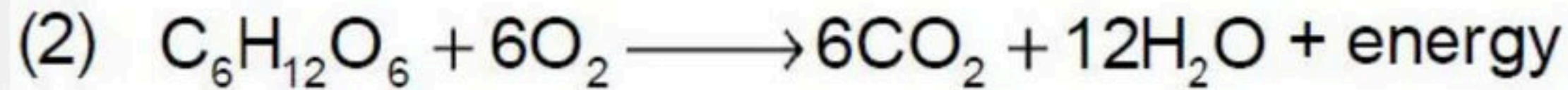
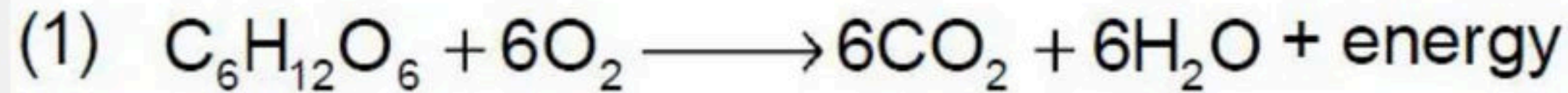


56. Photosynthesis is :

- (1) a physical process
- (2) a chemical process
- ✓ (3) a physio-chemical process
- (4) an energy wasting process



57 Which one is the correct summary equation of photosynthesis?



58. Which statement is correct :

I. Light reaction occurs in ~~stroma~~ grana

II. Light reaction occurs in grana and ATP and NADPH₂ are formed ✓

III. In stroma, dark reaction occurs ✓

IV. Dark reaction is not directly light driven but is dependent on the product (ATP and NADPH₂) formed in light reaction ✓

(1) All are correct

(2) All are incorrect

(3) II, III and IV are correct ✓

(4) I, II and IV are correct



59. Which one of the following is essential for photolysis of water :

- ☒ (1) Manganese
- (2) Zinc
- (3) Copper
- (4) Boron



60. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) 2025

Assertion (A) : Non cyclic photophosphorylation does not occur in stroma lamellae .

Reason (R) : In stroma lamellae membrane PS II and NADP reductase enzymes are not present.

choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- ☒ (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



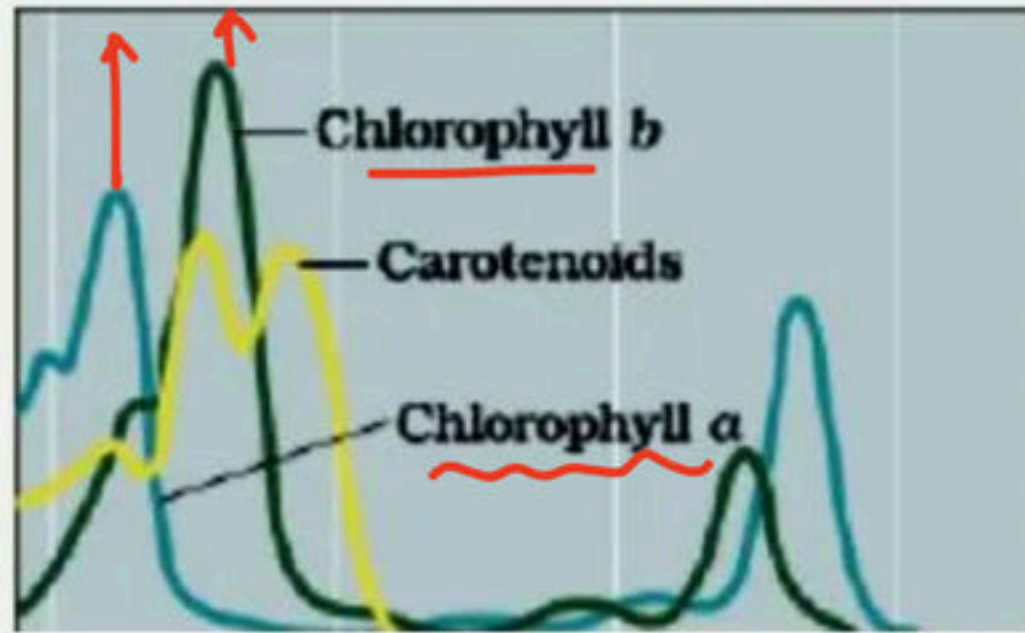
61. Which statement is not correct for photosynthesis reaction: Imp

- (1) H_2O is oxidised ✓
- (2) CO_2 is reduced ✓
- (3) It is a Anabolic reaction ✓
- (4) It is catabolic ✗



Incorrect

62. Which statement is not correct with above graph:



- (1) Chlorophyll b show greater absorption peak in blue light ✓
- ✓ (2) Chlorophyll a show greater absorption peak in red light and blue
- (3) In red light chl a have greater peak than chlorophyll b ✓
- (4) In blue light chl b have greater peak than chlorophyll a



63. Which event is not related with Z scheme :

- (1) Splitting of H_2O
- (2) $NADP^+$ reduction
- ~~(3) CO_2 reduction~~
- (4) ATP formation



64. Which of the following are produced by reactions that take place in the thylakoids and consumed by reactions in the stroma :

- (1) CO_2 and H_2O
- (2) NADP^+ and ADP
- ☒ (3) ATP and NADPH
- (4) Glucose and O_2



65. Conditions necessary for photosynthesis are :

- (1) Light and suitable temperature
- (2) Chlorophyll and water
- (3) Carbon dioxide
- (4) All of the above



66. Which of the following internal factors effects photosynthesis

(1) Temperature

(2) CO₂ concentration

✓ (3) Internal CO₂ concentration

(4) More than one correct

} external



67. Solarization is a process in which :

- (1) Sugars are formed with the help of solar energy
- (2) Chlorophyll is formed
- ✓ (3) Destruction of chlorophyll and ultimate death of living cell
- (4) Mobilization of light energy



V.V.IMP

68. In C₄ plants initial and final CO₂ fixation takes place in which cell respectively

(1) Bundle sheath cell and Mesophyll cell

O.A.A

✓ (2) Mesophyll cell and Bundle sheath cell

(3) Mesophyll cell and Guard cell

(4) Guard cell and Mesophyll cell



69. In C_4 plants

(1) Photorespiration takes place in mesophyll cells

(2) Photorespiration takes place in bundle sheath cells

✓ (3) Photorespiration do not takes place due to ↳ B.S.C
maintainance of high CO_2 concentration at enzyme active site

(4) Both 1 and 2



70. The correct sequential stages of calvin cycle are: 202 S

- ✓ (1) Carboxylation, Reduction and Regeneration
- (2) Carboxylation, Regeneration and Reduction
- (3) Reduction, Carboxylation and Regeneration
- (4) Regeneration, Reduction and Carboxylation



71. During initial carboxylation in C₄ plants the main enzyme required is :

- (1) RuBISCO
- ✓ (2) PEP-carboxylase
- (3) Decarboxylase
- (4) Phosphorylase



72. In ^{"PSII"} non-cyclic photophosphorylation ["] produces :

- (1) NADPH only
- (2) ATP and NADPH only
- ☒ (3) ATP, NADPH and O₂
- (4) ATP only



73. Lack of photorespiration gives which of the following benefits to C_4 plants

(1) higher productivity ✓

(2) higher yields ✓

✓ (3) both (1) and (2)

(4) Lower productivity



74. RuBisCO have higher affinity for

- ✓ (1) CO_2
- (2) O_2
- (3) both 1 and 2 with equal affinity
- (4) PGA



75. When Oxygen binds to RuBisCO enzyme the rate of CO_2 fixation

(1) Increases

✓ (2) Decreases

(3) Not effected

(4) Stops



76. How many Calvin cycles would generate one molecule of glucose/hexose

(1) One cycle

(2) Three cycle

✓ (3) Six cycles

(4) Twelve cycles



77. The first stable product of Calvin cycle and Hatch and Slack cycle are :

(1) 4-C and 3-C compounds

(2) 4-C and 6-C compounds

✓ (3) 3-C ^{→ PHA} and 4-C ^{→ OAA} compounds

(4) 5-C and 4-C compounds



78. In C_4 plants plasmodesmata are used for transport of

C_4A

(a) C_4 acid from mesophyll to bundle sheath cells ✓

(b) C_4 acid from bundle sheath to mesophyll cells ✗

(c) C_3 acid from mesophyll to bundle sheath cells ✗

(d) C_3 acid from bundle sheath cells to mesophyll cells

✓ (1) a and d PEP

(2) a and c

(3) b and d

(4) b and c



79. The correct sequence of flow of electrons in the light reaction is

- ✓ (1) PSII, plastoquinone, cytochromes, PSI, ferredoxin
- (2) PSI, plastoquinone, cytochromes, PSII, ferredoxin
- (3) PSI, ferredoxin, PSII,
- (4) PSI, plastoquinone, cytochromes, PSII, ferredoxin



80. Photorespiration is favoured by :

- (1) Low light intensity
- (2) Low O_2 and high CO_2
- (3) Low temperature
- ✓ (4) High O_2 and Low CO_2



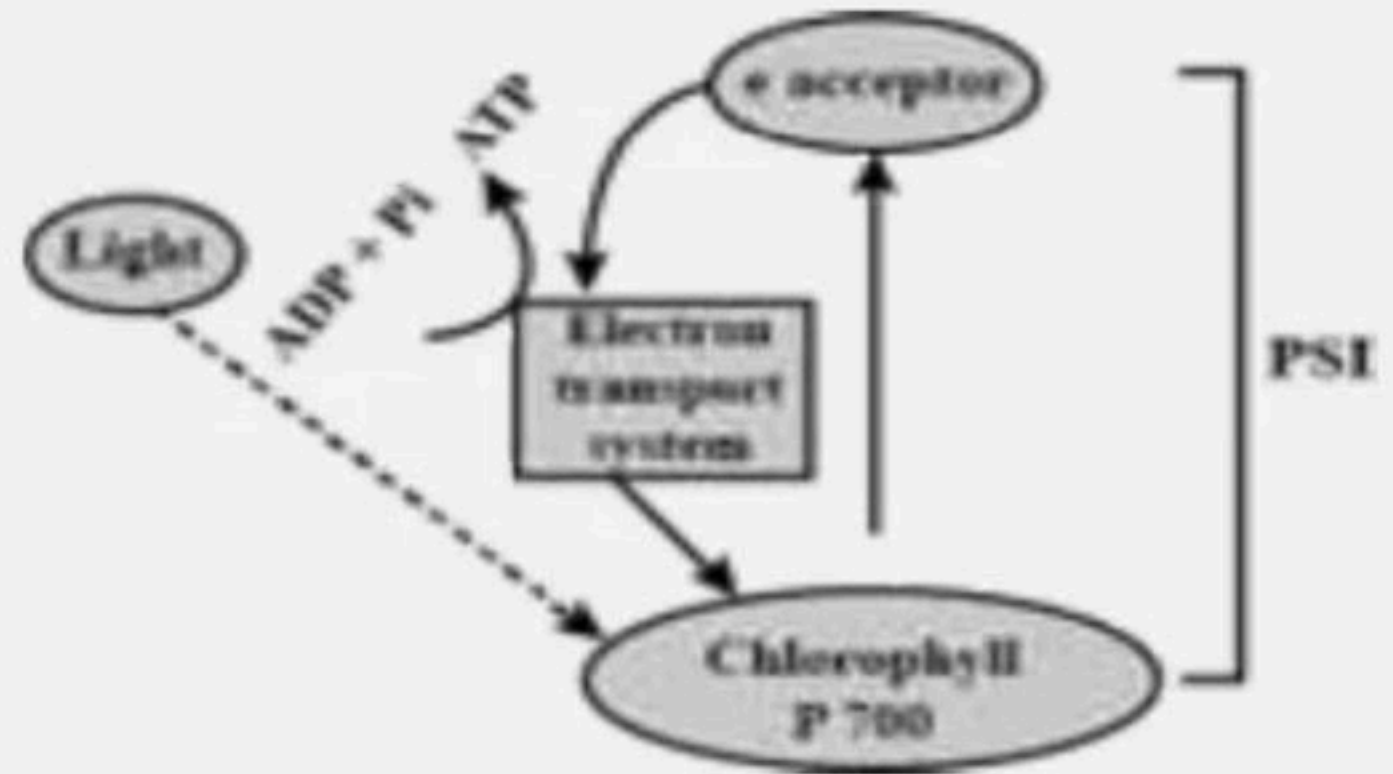
81. Which off the following is true During photosynthesis

- (1) Several factors interact and simultaneously effect photosynthesis ✓
- (2) Usually one factor is the major cause and limit the rate ✓
- (3) At any point the rate will be determined by the factor available at suboptimal levels ✓
- (4) All of these are true ✓



82. What does the given diagram represent with respect to the various photosynthetic process ?

- (1) C_2 cycle
- ✓ (2) Cyclic photophosphorylation
- (3) Non-Cyclic photophosphorylation
- (4) Z-scheme of phosphorylation



83. Major limiting factor for photosynthesis is

✓✓ (1) CO₂ ↩

(2) Light

(3) Temperature

(4) Water



84. CO_2 is necessary for photosynthesis : The chemical used to remove this gas most effectively from entering a control apparatus is :

- ✓ (1) KOH solution ↩
- (2) Calcium oxide
- (3) Sodium carbonate
- (4) Distilled water



85. Read the following four statements, A, B, C, and D, and select the right option having both correct statements:

A. Z scheme of light reaction takes place in presence of PS-I only. ✗

✓ B. Only PS-I is functional in cyclic photophosphorylation. ✓

C. Cyclic photophosphorylation results in synthesis of ATP and NADPH₂. ✗ ^{only}

✓ D. Stroma lamellae lack PS-II as well as NADP.

Options:

(1) A and B

(2) B and C

(3) C and D

(4) A and D

B and D



86. The following (P through U) are the main steps of chemosynthetic ATP synthesis in the light reaction.

Which answer places them in correct order?

P. H^+ concentration gradient generates (4)

Q. H^+ diffuses through ATP synthetase (5)

R. Carriers use energy from electrons to move H^+ across the membrane (3)

S. Electrons from PS-II pass along electron transport chain (2)

T. Light excites electrons in PS-II (1)

U. Energy of H^+ flow is used by ATP synthetase to make ATP (6)

Options:

(1) $P \rightarrow Q \rightarrow T \rightarrow S \rightarrow R \rightarrow U$

(2) $S \rightarrow T \rightarrow P \rightarrow Q \rightarrow R \rightarrow U$

(3) $T \rightarrow S \rightarrow R \rightarrow P \rightarrow Q \rightarrow U$

(4) $T \rightarrow S \rightarrow R \rightarrow U \rightarrow Q \rightarrow P$



87. Assuming a thylakoid is somehow punctured so that the interior of the thylakoid is no longer separated from the stroma. This damage will have the most direct effect on which of the following processes?

- (1) Splitting of water
- (2) Absorption of light energy by chlorophyll
- (3) Flow of electrons from photosystem II to photosystem I
- (4) Synthesis of ATP



88. The proton gradient created across the thylakoid membrane in chloroplast:

A. Increases pH of lumen ↓ X

B. Breaks due to movement of protons from lumen to stroma ✓

C. Breaks due to active transport of H⁺ through F₁ particle X (F₀)

The correct statements is/are:

(1) Only A

(2) Both B and C

(3) Only C

(4) Only B ✓



89. Algae employed by Calvin et al. in experiments on photosynthesis belong to: -

Exemplar

- (1) Euglena and Scenedesmus
- (2) Chara
- (3) Chlamydomonas^x and Chlorella
- (4) Chlorella and Scenedesmus



90. Which technique has helped in the investigation of Calvin cycle?

- (1) X-ray crystallography
- (2) X-ray technique
- ☒ (3) Radioactive isotope technique ✓
- (4) Intermittent light



91. How many ATP molecules are required to synthesize one molecule of glucose by C_4 pathways?

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2025

(1) 12

(2) 18

(3) 24

(4) 30



92. Photorespiration is favoured by-

- ✓ (1) High oxygen and low carbon dioxide
- (2) High carbon dioxide and low oxygen
- (3) High temperature and low oxygen
- (4) High humidity and temperature



93. The products of photochemical reaction are

- ✓ (1) O₂, ATP, and NADPH ✓
- (2) O₂ only
- (3) ATP and NADPH
- (4) Organic compounds especially carbohydrates



94. Electrons from excited chlorophyll of PS II are accepted first by

- ✓ (1) Quinone
- (2) Ferredoxin
- (3) Cytochrome bbb
- (4) Cytochrome fff



95. Photolysis of 2 water molecules yields

(1) 2 electrons and 4 protons

✓ (2) 4 electrons and 4 protons

(3) 4 electrons and 2 protons

(4) 2 electrons and 2 protons



96. ADP $\rightarrow$ ATP reaction occurs when protons (H^+) are passed from

- (1) Thylakoid lumen to cytosol
- (2) Thylakoid to lamella
- ✓ (3) Lumen of thylakoid to stroma
- (4) Stroma to thylakoid lumen



97. Synthesis of ADP + iP $\rightarrow$ ATP in grana/photosynthesis is

(1) Phosphorylation (ATP synthase)

✓ (2) Photophosphorylation

(3) Oxidative phosphorylation

(4) Photolysis



98. Which one is important in the electron transport system?

- ✓ (1) Ubiquinone
- (2) Acetyl CoA
- (3) Tricarboxylic acid
- (4) Ferricyanide



99. Which of the following is the most common H⁺ acceptor?

- (1) FAD and NAD⁺
- (2) FMN and NAD⁺
- (3) FMN and FAD
- (4) NAD⁺ and NADP⁺

V.V.V.V.V. Imp



100. Which one directly transfers electrons to NADP⁺ during the light reaction?

- ✓ (1) PS I ←
- (2) PS II
- (3) Cytochromes
- (4) Plastocyanin



THANK YOU



BIOLOGYLIVE



Thank You

NEET 2024 Questions In Test Series

NEET 2024.



Test Series



71. Match List I with List II

List I
(Quantum Number)

- A. m_l
- B. m_s
- C. l
- D. n

List II
(Information provided)

- I. Shape of orbital
- II. Size of orbital
- III. Orientation of orbital
- IV. Orientation of spin of electron

Choose the correct answer from the options given below :

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B-I, C-IV, D-III

Answer (2)

Question-92 Match the following-

List - A

List -B

- | | |
|----------|------------------------------------|
| a. n | 1. size of orbital |
| b. l | 2. orientation of spin of electron |
| c. m | 3. shape of the orbital |
| d. m_s | 4. orientation of the orbital |

A , b ,c , d respectively are

Type	Multiple choice	
Option	2 ,1 ,4 ,3	incorrect
Option	1 ,2 ,3 ,4	incorrect
Option	1 ,3 ,4 ,2	correct
Option	3 ,4 ,2 ,1	Incorrect
Solution		
Marks	4	1

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61. Activation energy of any chemical reaction can be calculated if one knows the value of
- (1) rate constant at standard temperature
 - (2) probability of collision
 - (3) orientation of reactant molecules during collision
 - (4) rate constant at two different temperatures

Answer (4)

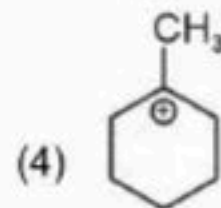
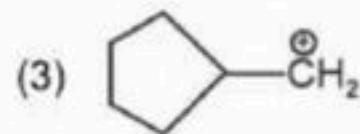
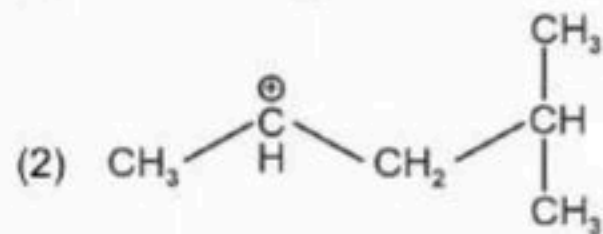
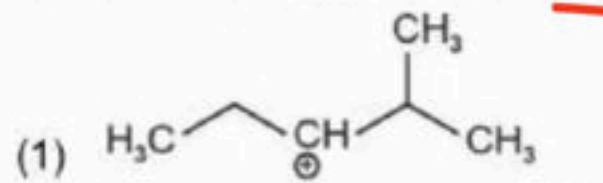
Test Series

Question-96	Activation energy of a chemical reaction can be determined by	
Type	Multiple_choice	
Option	Determining the rate constant at standard temp	incorrect
Option	Determining the rate constants at two temperatures	Correct
Option	Using catalyst	incorrect
Option	Determining probability of collision	Incorrect
Solution		
Marks	4	1



NEET 2024.

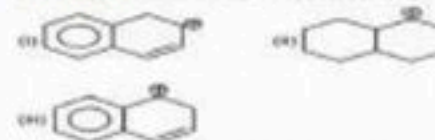
60. The most stable carbocation among the following is :



Answer (4)

Test Series

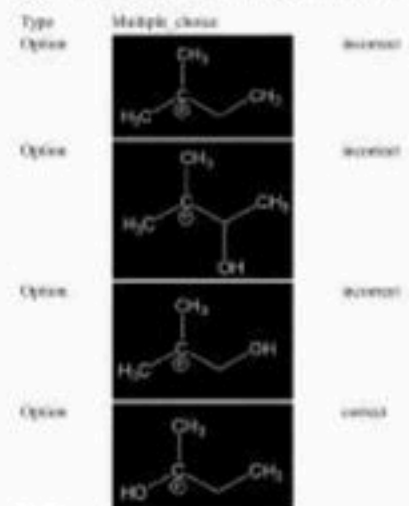
Question 60 Consider the following Carbocations:



The correct order of their stability is?

Type: Multiple choice
Options: (1) < (2) < (3), (2) < (1) < (3), (2) < (3) < (1), (1) < (3) < (2)
Solution: correct
Marks: 4

Question 75 Which of the following cations is most stable?

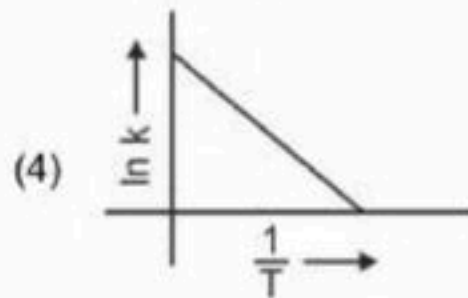
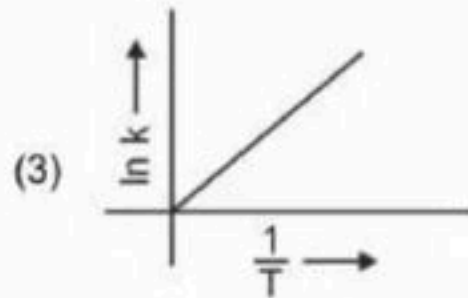
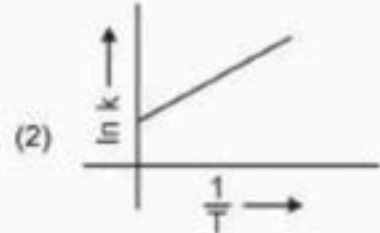
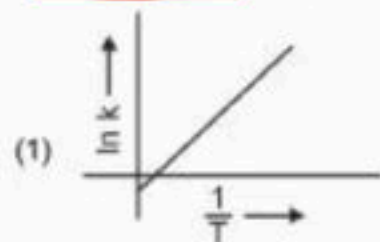


There are more than 20+ different question on stability of carbonation in Test series

NEET 2024.

Test Series

67. Which plot of $\ln k$ vs $\frac{1}{T}$ is consistent with Arrhenius equation?



Answer (4)

2 PM

94
100

195KA



4992 ←
PandC

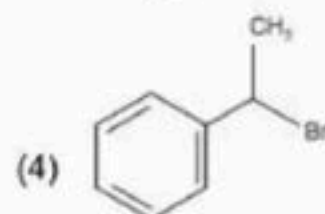
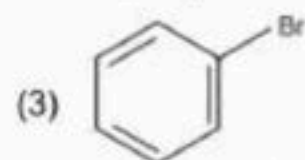
SECTION-B

Question-86	For Arrhenius equation, Which among following is a correct plot?	
Type	Multiple choice	
Option		incorrect
Option		correct
Option		incorrect
Option		incorrect
Solution		
Marks	4	1

1st feb - 11 PM 1P

NEET 2024.

56. The compound that will undergo S_N1 reaction with the fastest rate is



Answer (4)

Test Series

Question-72 Which of the following is most reactive for S_N1 :

Type Multiple_choice

Option $(CH_3)_3C-Cl$

correct

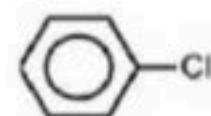
Option $CH_2=CHCl$

incorrect

Option CH_3Cl

incorrect

Option



incorrect

Solution Reactivity for S_N1 reaction is directly proportional to stability of carbocation.

Marks

4

1

There are 10+ different question on S_N1 in Test series

NEET 2024.

Test Series

63. Arrange the following elements in increasing order of electronegativity:

N, O, F, C, Si

Choose the correct answer from the options given below:

(1) $\text{Si} < \text{C} < \text{N} < \text{O} < \text{F}$

(2) $\text{Si} < \text{C} < \text{O} < \text{N} < \text{F}$

(3) $\text{O} < \text{F} < \text{N} < \text{C} < \text{Si}$

(4) $\text{F} < \text{O} < \text{N} < \text{C} < \text{Si}$

Answer (1)

Question-91 The electronegativity values of C,N,O and F :

Type Multiple choice

Option increase from carbon to fluorine.

correct

Option decrease from carbon to fluorine.

incorrect

Option increase up to oxygen and is minimum at fluorine.

incorrect

Option is minimum at nitrogen and then increase continuously.

incorrect

Solution As size of atom decreases across the period, the attraction between the nucleus and shared pair of electrons increases. So electronegativity increases across the period.

Marks

4

1



NEET 2024.

52. The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order:

- (1) $B > A > C$
- (2) $B > C > A$
- (3) $A > C > B$
- (4) $A > B > C$

Answer (2)

Test Series



Question-57 For different gases the correct statement about Henry's constant (K_H) is-

Type	Multiple_choice	
Option	higher the value of K_H at a given pressure, higher is the solubility of the gas	Incorrect
Option	higher the value of K_H at a given pressure, lower is the solubility of the gas	Correct
Option	K_H is not a function of nature of gas	Incorrect
Option	K_H value for all gases is same at a given pressure	Incorrect
Solution Marks	4	1



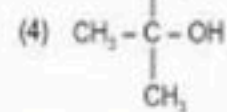
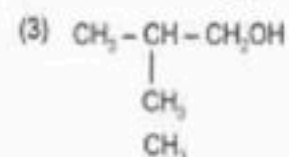
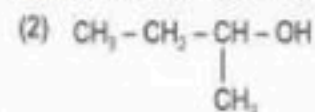
This is a question you'll rarely found in any test series

This is what we provide line to line from Ncert

NEET 2024.

Test Series

57. Which one of the following alcohols reacts instantaneously with Lucas reagent?



Answer (4)

Question-85 **Assertion :** In Lucas test, 3^o alcohols react immediately.
Reason : An equimolar mixture of anhyd. ZnCl_2 and conc. HCl is called Lucas reagent

Type Multiple_choice

Option If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion incorrect

Option If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion. correct

Option If the Assertion is correct but Reason is incorrect. incorrect

Option If both the Assertion and Reason are incorrect. incorrect

Solution

Marks 4 1



Same concept



NEET 2024.

Test Series

88. Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is (Given : Molar mass of Cu : 63 g mol⁻¹, 1 F = 96487 C)

- (1) 3.15 g
- (2) 0.315 g
- (3) 31.5 g
- (4) 0.0315 g

Answer (2)

Question-72	How much metal will be deposited when a current of 12 ampere with 75% efficiency is passed through the cell for 3 h? (Given: $Z = 4 \times 10^{-4}$)	
Type	Multiple_choice	
Option	32.4 g	incorrect
Option	38.8 g	correct
Option	36.0 g	incorrect
Option	22.4 g	incorrect
Solution	$W = Z \cdot I \cdot t$ $= 4 \times 10^{-4} \times 12 \times 3 \times 3600 = 38.8 \text{ g}$	
Marks	4	1



Same concept



NEET 2024.

Test Series

55. Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order

n -pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

Answer (1)

Question-81

B.P. of branched chain alkanes as compared to straight chain alkanes are

Type Multiple_choice

Option Lower

correct

Option Equal

incorrect

Option Higher

incorrect

Option Independent of the chain

incorrect

Solution As the number of branches increases, surface area decreases, due to which Vander Waal forces of attraction decreases. Hence, boiling point also decreases

Marks 4

1

Question-79

Which of the following has maximum boiling point

Type Multiple_choice

Option *iso*-octane

incorrect

Option *n*-octane

correct

Option 2, 2, 3, 3-tetramethyl butane

incorrect

Option *n*-butane

incorrect

Solution *n*-octane

Boiling point depends on molecular mass. Greater the molecular mass higher will be the boiling point.

Boiling point also depends on the structure. If two compounds have same molecular mass then straight chain or linear compound has higher boiling point.

Marks 4

1

Many Repetitive questions are practising like this in our test series that directly comes in neet



Same concept



NEET 2024.

Test Series

80. Fehling's solution 'A' is

- (1) aqueous copper sulphate
- (2) alkaline copper sulphate
- (3) alkaline solution of sodium potassium tartrate (Rochelle's salt)
- (4) aqueous sodium citrate

Answer (1)

Question-53 Fehling's Solution is

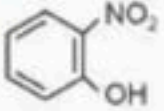
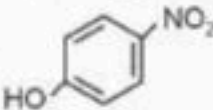
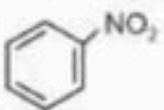
Type	Multiple_choice	
Option	Amonical Cuprous chloride	incorrect
Option	Aqueous copper sulphate	correct
Option	AgNo ₃ solution	InCorrect
Option	Sodium chloride	incorrect
Solution		
Marks	4	1



NEET 2024.

Test Series

54. Intramolecular hydrogen bonding is present in

- (1) 
- (2) 
- (3) 
- (4) HF

Answer (1)

Question-93 Which one of the following does not have intermolecular H-bonding ?

Type Multiple choice

Option H_2O correct

Option o-nitro phenol incorrect

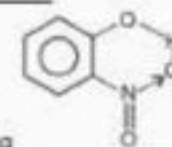
Option HF incorrect

Option CH_3COOH incorrect

Solution

It has intramolecular H-bonding

Marks 4 1



NEET 2024.

Test Series

59. Arrange the following elements in increasing order of first ionization enthalpy:

Li, Be, B, C, N

Chose the correct answer from the options given below:

- (1) $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{N}$
- (2) $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N}$
- (3) $\text{Li} < \text{Be} < \text{C} < \text{B} < \text{N}$
- (4) $\text{Li} < \text{Be} < \text{N} < \text{B} < \text{C}$

Answer (2)

Question-66 Correct order of 1st ionization potential among elements Be, B, C, N, O is

Type Multiple_choice

Option $\text{B} < \text{Be} < \text{C} < \text{O} < \text{N}$

correct

Option $\text{B} < \text{Be} < \text{C} < \text{N} < \text{O}$

incorrect

Option $\text{Be} < \text{B} < \text{C} < \text{N} < \text{O}$

incorrect

Option $\text{Be} < \text{B} < \text{C} < \text{O} < \text{N}$

Incorrect

Solution NA

Marks 4

1



NEET 2024.

100. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :

(Given atomic masses of A = 64; B = 40; C = 32 u)

- (1) A_2BC_2
- (2) ABC_3
- (3) AB_2C_2
- (4) ABC_4

Answer (2)

Test Series

Question-94 An organic compound containing C, H and N gave the following analysis C = 40%, H = 13.33%, N = 46.47%. Its empirical formula would be

Type Multiple_choice

Option CH_4N correct

Option CH_3N incorrect

Option $C_2H_7N_2$ incorrect

Option C_2H_7N incorrect

Solution CH_4N

Marks 4

1

□

Question-87 A compound contains 54.55% carbon, 9.09% hydrogen and 36.36% oxygen. The empirical formula of this compound is-

Type Multiple_choice

Option C_7H_4O incorrect

Option $C_3H_4O_2$ incorrect

Option $C_2H_4O_2$ incorrect

Option C_2H_2O correct

Solution C_2H_2O

Marks 4

1



Same

concept



NEET 2024.

73. The highest number of helium atoms is in

- (1) 4 mol of helium
- (2) 4 u of helium
- (3) 4 g of helium
- (4) 2.271098 L of helium at STP

Answer (1)

Test Series

Question-95 Rearrange the following in the order of increasing masses:

- (i) 0.5 mole of O_3
- (ii) 0.5 gm atom of oxygen
- (iii) 3.011×10^{23} molecules of O_2
- (iv) 5.6 litre of CO_2 at STP

Type	Multiple_choice	
Option	ii < iv < iii < i	correct
Option	ii > iv > iii > i	incorrect
Option	i < iii < ii < iv	incorrect
Option	i > iii > ii > iv	incorrect
Solution	ii < iv < iii < i	
Marks	4	1

Question-55 Which one has maximum number of atoms ?

Type	Multiple_choice	
Option	0.25 mole NH_4NO_3	InCorrect
Option	40 gram CH_4	correct
Option	40 gm helium	Incorrect
Option	36 gram glucose	incorrect
Solution		
Marks	4	1

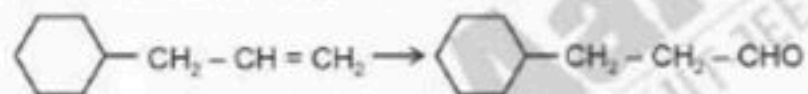


Same concept



NEET 2024.

77. Identify the correct reagents that would bring about the following transformation.

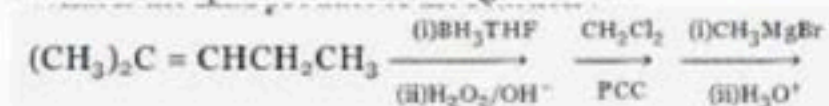


- (1) (i) $\text{H}_2\text{O}/\text{H}^+$
 (ii) CrO_3
- (2) (i) BH_3
 (ii) $\text{H}_2\text{O}_2 / \text{OH}^-$
 (iii) PCC
- (3) (i) BH_3
 (ii) $\text{H}_2\text{O}_2 / \text{OH}^-$
 (iii) alk. KMnO_4
 (iv) H_3O^+
- (4) (i) $\text{H}_2\text{O}/\text{H}^+$
 (ii) PCC

Answer (2)

Test Series

Question-69 Find the final product of following reaction :



Type

Multiple_choice

Option

2,4-dimethyl-3-pentanol

incorrect

Option

2,3-dimethyl-3-pentanol

incorrect

Option

2,2-dimethyl-2-pentanol

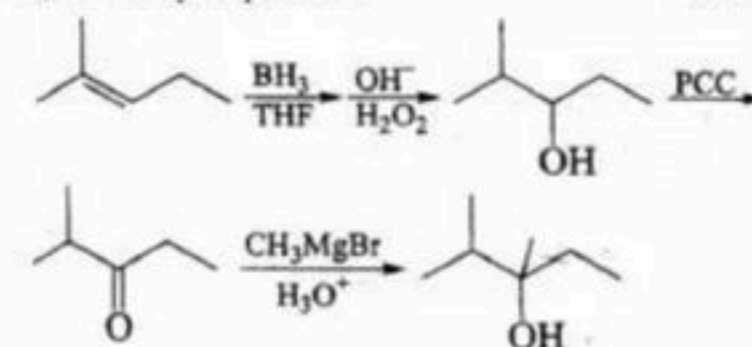
incorrect

Option

2,3-dimethyl-3-pentanol

correct

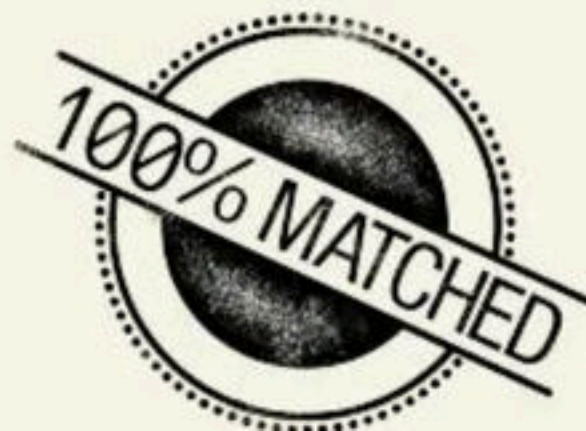
Solution



Marks

4

1



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Test Series

68. The energy of an electron in the ground state ($n = 1$) for He^+ ion is $-x$ J, then that for an electron in $n = 2$ state for Be^{3+} ion in J is

- (1) $-x$
- (2) $-\frac{x}{9}$
- (3) $-4x$
- (4) $-\frac{4}{9}x$

Answer (1)

Question-60 Energy of an electron in a one-electron system can be calculated as

$$E_n = \frac{-2.18 \times 10^{-18} Z^2}{n^2}$$

Which of the following correctly states the relationship between the $n=2$ level of He^+ ion ($Z = 2$) and $n=2$ level of Li^{2+} ion ($Z = 3$) ?

Type Multiple choice

Option $E_{\text{He}^+} = 9/4 E_{\text{Li}^{2+}}$ Incorrect

Option $E_{\text{He}^+} = 4/9 E_{\text{Li}^{2+}}$ correct

Option $E_{\text{He}^+} = 9/2 E_{\text{Li}^{2+}}$ Incorrect

Option $E_{\text{He}^+} = 2/9 E_{\text{Li}^{2+}}$ InCorrect

Solution

Marks 4

1



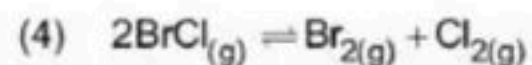
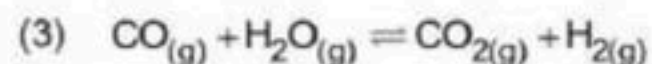
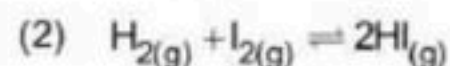
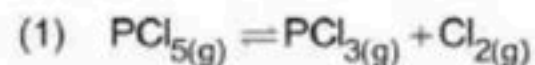
Same concept



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Test Series

78. In which of the following equilibria, K_p and K_c are **NOT** equal?



Answer (1)

Question-56 For which reaction is $K_p = K_c$:

Type Multiple_choice

Option $2\text{NOCl}(g) \rightleftharpoons 2\text{NO}(g) + \text{Cl}_2(g)$ correct

Option $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$ Incorrect

Option $\text{H}_2(g) + \text{Cl}_2(g) \rightleftharpoons 2\text{HCl}(g)$ Incorrect

Option $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g)$ incorrect

Solution for $K_p = K_c$, $\Delta n = 0$
only option (3) with satisfy this condition.

Marks 4 1



Same concept



NEET 2024.

Test Series

98. The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is

(Given $R = 2.0 \text{ cal K}^{-1} \text{ mol}^{-1}$)

(1) 0 calorie

(2) -413.14 calories

(3) 413.14 calories

(4) 100 calories

Answer (2)

Question-53 Calculate the change in entropy when 1 mole gas expands isothermally and reversibly from an initial volume of 1 L to a final volume of 10 L at 27°C:

Type Multiple_choice

Option 14.47 J K⁻¹

incorrect

Option 19.12 J K⁻¹

correct

Option 21.98 J K⁻¹

incorrect

Option 38.6 J K⁻¹

incorrect

Solution 19.12 J K⁻¹

Marks 4

1

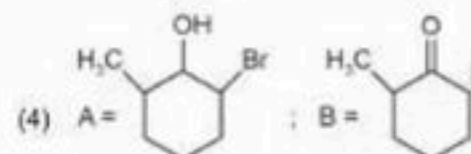
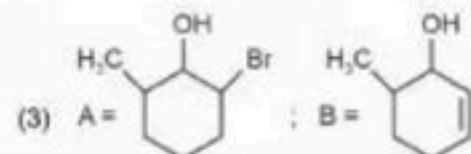
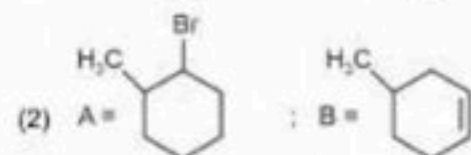
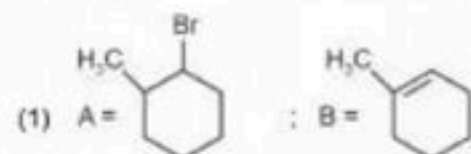
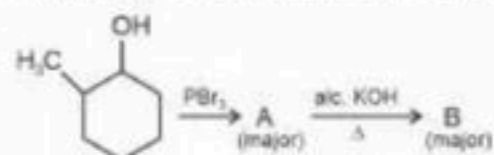


Same concept



NEET 2024.

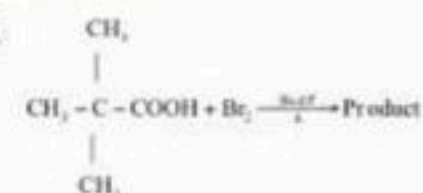
93. Major products A and B formed in the following reaction sequence, are



Answer (1)

Test Series

Question-93



The product of the above reaction is -

Type

Multiple choice

Option

β -Dibromo acid

incorrect

Option

β, β' - Dibromo acid

incorrect

Option

β, β', β'' -Tribromo acid

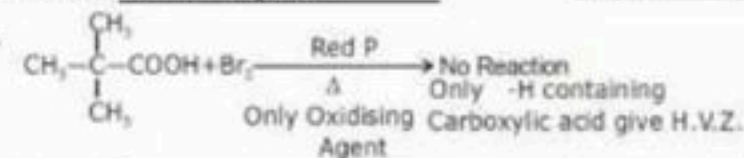
incorrect

Option

No reaction takes place

correct

Solution



Marks

4

1



Same

concept



NEET 2024.

Test Series

81. Match List I with List II.

List-I

(Process)

- A. Isothermal process
- B. Isochoric process
- C. Isobaric process
- D. Adiabatic process

List-II

(Conditions)

- I. No heat exchange
- II. Carried out at constant temperature
- III. Carried out at constant volume
- IV. Carried out at constant pressure

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-IV, B-II, C-III, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

Answer (4)

Question-75 Match the following :

- A- isothermal P - $Q = 0$
- B- Isobaric Q - Volume constant
- C- Isochoric R- Pressure Constant
- D- Adiabatic S- Temp. Constant

Type Multiple_choice

Option A - S, B - R, C - Q, D - correct P,

Option A - P, B - S, C - R, D - Incorrect Q,

Option A - R, B - P, C - Q, D - incorrect S,

Option A - Q, B - R, C - P, D - incorrect S,

Solution

Marks 4

1



NEET 2024.

69. In which of the following processes entropy increases?

- A. A liquid evaporates to vapour.
- B. Temperature of a crystalline solid lowered from 130 K to 0 K.
- C. $2\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- D. $\text{Cl}_{2(\text{g})} \rightarrow 2\text{Cl}_{(\text{g})}$

Choose the correct answer from the options given below:

- (1) A and C
- (2) A, B and D
- (3) A, C and D
- (4) C and D

Answer (3)

Test Series

Question-82 Which one of the following has ΔS° greater than zero?

Type Multiple choice

Option $\text{CaO}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons \text{CaCO}_3(\text{s})$

incorrect

Option $\text{NaCl}(\text{aq}) \rightleftharpoons \text{NaCl}(\text{s})$

incorrect

Option $\text{NaNO}_3(\text{s}) \rightleftharpoons \text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$

correct

Option $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

incorrect

Solution An entropy (ΔS) of a reaction is positive if the products are in more random state as compared to reactants (order of randomness is gas > liquid > solid).

$\text{NaNO}_3(\text{s}) \rightleftharpoons \text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$

This process is accompanied by increase in randomness because Na^+ and NO_3^- ions are free to move in solution whereas they are not free to move in solid NaNO_3 .

In (1) product (solid) has lesser entropy as compared to reactants.

In (2) $\text{NaCl}(\text{solid})$ has lesser entropy as compared to $\text{NaCl}(\text{aq})$ (Na^+ and Cl^-).

In (4) both reactants and products are in gaseous state but number of moles of products are decreasing.

Marks

4

1



Same

concept



NEET 2024.

85. For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is: $[A] = [B] = [C] = 2 \times 10^{-3} \text{ M}$.

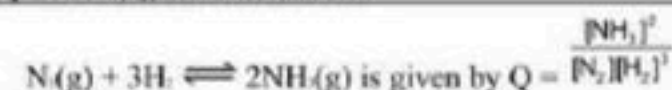
Then, which of the following is correct?

- (1) Reaction is at equilibrium.
- (2) Reaction has a tendency to go in forward direction.
- (3) Reaction has a tendency to go in backward direction.
- (4) Reaction has gone to completion in forward direction.

Answer (3)

Test Series

Question-95 The reaction quotient (Q) for the reaction



The reaction will proceed from right to left if

Type Multiple choice

Option $Q = K_c$ incorrect

Option $Q < K_c$ incorrect

Option $Q > K_c$ correct

Option $Q = 0$ incorrect

Solution For reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$Q \text{ (Quotient)} = \frac{[NH_3]^2}{[N_2][H_2]^3}, \quad \Delta N = 2 - 4 = -2$$

At equilibrium Q is equal to K, but for the progress of reaction towards right side $Q > K$.

Marks 4 1

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Test Series

62. Given below are two statements:

Statement I : Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is correct but Statement II is false
- (4) Statement I is incorrect but Statement II is true

Answer (1)

Question-63 Amine that cannot be prepared by Gabriel-phthalimide synthesis is:

Type Multiple_choice

Option Aniline correct

Option Benzyl amine incorrect

Option Methyl amine incorrect

Option Iso-butylamine incorrect

Solution Aniline

Marks 4 1



Same

concept



NEET 2024.

Test Series

74. Among Group 16 elements, which one does **NOT** show -2 oxidation state?

- (1) O
- (2) Se
- (3) Te
- (4) Po

Answer (4)

Question-95 The correct increasing order of stability of $+1$ oxidation state :

Type	Multiple choice	
Option	Tl > In > Ga > Al	incorrect
Option	Ga < In < Al < Tl	incorrect
Option	Al < Ga < In < Tl	correct
Option	Tl < In < Ga < Al	incorrect
Solution		
Marks	4	1



Same concept



NEET 2024.

Test Series

76. The reagents with which glucose does **not** react to give the corresponding tests/products are

- A. Tollen's reagent
- B. Schiff's reagent
- C. HCN
- D. NH_2OH
- E. NaHSO_3

Choose the correct options from the given below:

- (1) B and C
- (2) A and D
- (3) B and E
- (4) E and D

Answer (3)

499 = X

6 PM

299

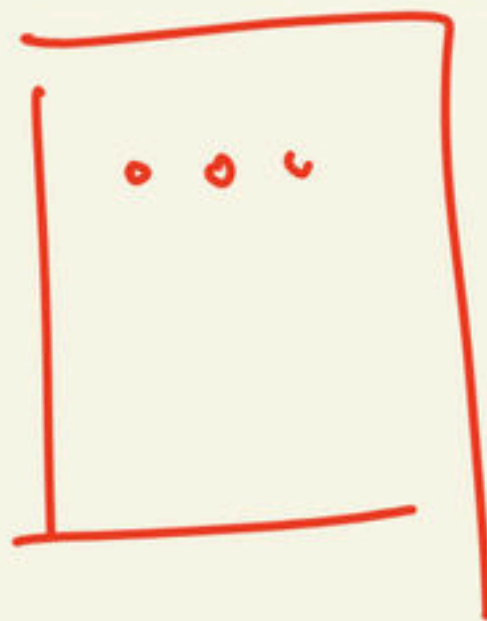
299

3002

3002

6002

60 + 300



Same concept



Question-74 Which of following is / are correct ?

(A) Glucose gives Fehling solution test but not gives Schiff's reagent test.

(B) Hydrolysis of protein gives α -amino acid

(C) Sucralose is derivative of sucrose but do not have glucose ring

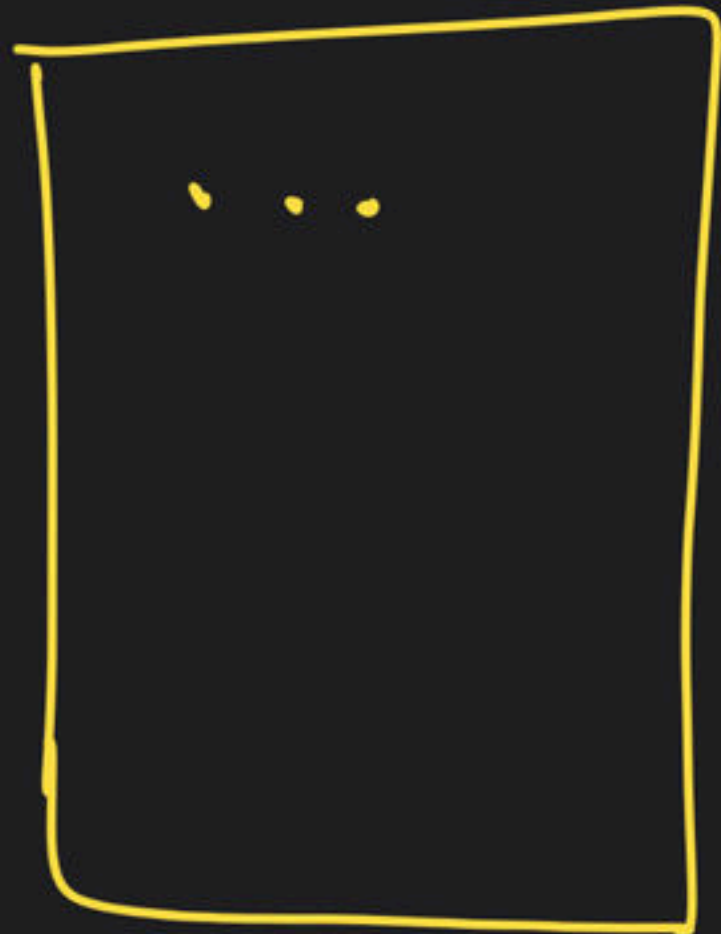
Type Multiple_choice

Option A, B	Incorrect
Option B,C	Incorrect
Option Only C	InCorrect
Option A, B, C	correct

Solution Marks 4 1

O.P

==



Allen → SRG



Non



NEET 2024.

Test Series

90. The products A and B obtained in the following reactions, respectively, are



(1) POCl_3 and H_3PO_3

(2) POCl_3 and H_3PO_4

(3) H_3PO_4 and POCl_3

(4) H_3PO_3 and POCl_3

Answer (4)



Question-98 The products formed by complete hydrolysis of PCl_3 are:

Type	Multiple choice	
Option	H_3PO_3 and HCl	correct
Option	POCl_3 and HCl	InCorrect
Option	H_3PO_4 and HCl	Incorrect
Option	$\text{H}_4\text{P}_2\text{O}_7$ and HCl	InCorrect
Solution		
Marks	4	1



NEET 2024.

Test Series

64. Match List I with List II.

List I (Conversion)	List II (Number of Faraday required)
A. 1 mol of H_2O to O_2	I. 3F
B. 1 mol of MnO_4^- to Mn^{2+}	II. 2F
C. 1.5 mol of Ca from molten CaCl_2	III. 1F
D. 1 mol of FeO to Fe_2O_3	IV. 5F

Choose the correct answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
 (2) A-III, B-IV, C-I, D-II
 (3) A-II, B-III, C-I, D-IV
 (4) A-III, B-IV, C-II, D-I

Answer (1)

Question-51 In acidic medium MnO_4^- is converted to Mn^{+2} when acts as an oxidising agent. The quantity of electricity required to reduce 0.05 mole of MnO_4^- would be -

Type Multiple_choice

Option 0.01 F incorrect

Option 0.05 F incorrect

Option 0.25 F correct

Option 0.15 F incorrect

Solution

Marks 4 1



Same concept



NEET 2024.

Test Series

51. Match List I with List II.

List I

(Molecule)

- A. ethane
- B. ethene
- C. carbon molecule, C_{60}
- D. ethyne

List II

(Number and types of bonds between two carbon atoms)

- I. one σ -bond and two π -bonds
- II. two π -bonds
- III. one σ -bonds
- IV. one σ -bond and one π -bond

Choose the correct answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-IV, C-I, D-II

Answer (3)

Question-74 The enolic form of ethyl acetoacetate has

Type Multiple_choice

- | | | |
|--------|-------------------------------|-----------|
| Option | 9 sigma bonds and 1 pi-bond | InCorrect |
| Option | 18 sigma bonds and 2 pi-bonds | correct |
| Option | 16 sigma bonds and 1 pi-bond | Incorrect |
| Option | 9 sigma bonds and 2 pi-bonds | Incorrect |

Solution

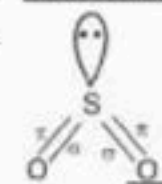
Marks 4 1

Question-73 Number of bonds in SO_2 are :

Type Multiple_choice

- | | | |
|--------|--|-----------|
| Option | two σ and two π | incorrect |
| Option | two σ and one π | incorrect |
| Option | two σ , two π and one lone pair | correct |
| Option | none of these | incorrect |

Solution



Marks 4 1



Same

concept



NEET 2024.

Test Series

84. Match List I with List II.

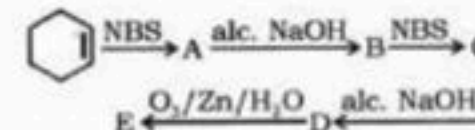
List I (Reaction)	List II (Reagents/Condition)
A. <chem>C1CCC(CC1)=C2CCCCC2 >> 2 C1CCC(=O)CC1</chem>	I. <chem>c1ccccc1C(=O)Cl</chem> /Anhyd. <chem>AlCl3</chem>
B. <chem>c1ccccc1 >> c1ccccc1C(=O)c2ccccc2</chem>	II. <chem>CrO3</chem>
C. <chem>C1CCC(CC1)O >> C1CCC(=O)CC1</chem>	III. <chem>KMnO4/KOH, \Delta</chem>
D. <chem>c1ccccc1CC >> c1ccccc1C(=O)[O-][K+]</chem>	IV. (i) <chem>O3</chem> (ii) <chem>Zn-H2O</chem>

Choose the correct answer from the options given below:

- (1) A-IV, B-I, C-III, D-II
- (2) A-III, B-I, C-II, D-IV
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-IV, C-II, D-III

Answer (3)

Question-82



Type	Multiple choice	
Option	<chem>CHO</chem> <chem>CHO</chem> <chem>CH2OH</chem> <chem>CH2OH</chem>	correct
Option	<chem>COOH</chem> <chem>COOH</chem> <chem>CH2COOH</chem>	incorrect
Option		incorrect
Option		incorrect
Solution		
Marks	4	1

Question-89 Dilute aqueous KMnO4, at room temperature reacts with R-CH=CH-R to give

Type	Multiple choice	
Option	<chem>R-CHO</chem>	incorrect
Option	<chem>R-COOH</chem>	incorrect
Option	<chem>RCHOH-CHOHR</chem>	correct
Option	<chem>CO2 + H2O</chem>	incorrect
Solution	$\text{R}-\text{CH}=\text{CH}-\text{R} \xrightarrow[\text{room temp.}]{\text{dilute aq. KMnO}_4} \text{R}-\underset{\text{OH}}{\text{CH}}-\underset{\text{OH}}{\text{CH}}-\text{R}$ $\text{R}-\text{CH}=\text{CH}-\text{R} \xrightarrow[\text{heat}]{\text{conc. KMnO}_4} \text{R}-\text{COOH} + \text{R}-\text{COOH}$	
Marks	4	1



Same

concept



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Test Series

53. Given below are two statements:

Statement I: The boiling point of hydrides of Group 16 elements follow the order

$\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$.

Statement II: On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O , it has higher boiling point.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

Answer (1)

Question-78 The decreasing order of the boiling points of the following hydrides is:

Type Multiple choice

Option $\text{H}_2\text{O} > \text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$

Incorrect

Option $\text{H}_2\text{O} > \text{NH}_3 > \text{SbH}_3 > \text{AsH}_3 > \text{PH}_3$

Correct

Option $\text{H}_2\text{O} > \text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$

Incorrect

Option $\text{H}_2\text{O} > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{NH}_3$

Incorrect

Solution

Marks 4

1



NEET 2024.

Test Series

91. The rate of a reaction quadruples when temperature changes from 27°C to 57°C . Calculate the energy of activation.

Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 4 = 0.6021$

- (1) 38.04 kJ/mol
- (2) 380.4 kJ/mol
- (3) 3.80 kJ/mol
- (4) 3804 kJ/mol

Answer (1)

Question-98 The rate constant, the activation energy and the arrhenius parameter of a chemical reaction at 25°C are $3.0 \times 10^{-4} \text{ s}^{-1}$, $104.4 \text{ kJ mol}^{-1}$ and $6.0 \times 10^{14} \text{ s}^{-1}$ respectively. The value of the rate constant as $T \rightarrow \infty$ is

Type Multiple_choice

Option $2.0 \times 10^{18} \text{ s}^{-1}$ incorrect

Option $6.0 \times 10^{14} \text{ s}^{-1}$ correct

Option Infinity incorrect

Option $3.6 \times 10^{30} \text{ s}^{-1}$ incorrect

Solution $T_2 = T(\text{say}), T = 25^{\circ}\text{C} = 298 \text{ K}$.

$$E_a = 104.4 \text{ kJ mol}^{-1} = 104.4 \times 10^3 \text{ J mol}^{-1}$$

$$K_1 = 3 \times 10^{-4}, K_2 = ?$$

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$



Same

concept



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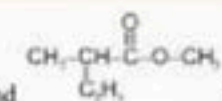
Test Series

79. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is :

- (1) n-hexane
- (2) 2-methylpentane
- (3) 2,3-dimethylbutane
- (4) 2,2-dimethylbutane

Answer (3)

Question-74



The correct IUPAC name of compound is :

Type	Multiple choice
Option	Methyl 2-ethylpropanoate incorrect
Option	Methyl butane-2-carboxylate incorrect
Option	Methyl 2-methylbutanoate correct
Option	Methoxypentanone incorrect

Solution

$$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3-\text{CH}-\text{C}-\text{O}-\text{CH}_3 \\ | \\ \text{CH}_2-\text{CH}_3 \\ 3 \quad 4 \end{array}$$

Methyl 2-methylbutanoate

Marks 4 1

More than 15 question in test series of nomenclature for practice well

NEET 2024.

95. Identify the **correct** answer.
- (1) Three resonance structures can be drawn for ozone
 - (2) BF_3 has non-zero dipole moment
 - (3) Dipole moment of NF_3 is greater than that of NH_3
 - (4) Three canonical forms can be drawn for CO_3^{2-} ion

Answer (4)

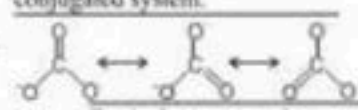


Test Series

Question-87 The correct order of dipole moment is :

Type	Multiple choice
Option	$\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$ correct
Option	$\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$ incorrect
Option	$\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$ incorrect
Option	$\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$ incorrect
Solution	$\text{H}_2\text{O}, \mu = 6.17 \times 10^{-30} \text{ Cm}; \text{NH}_3, \mu = 4.90 \times 10^{-30} \text{ Cm};$ $\text{NF}_3, \mu = 0.80 \times 10^{-30} \text{ Cm}; \text{CH}_4, \mu = \text{zero}.$
Marks	4 1

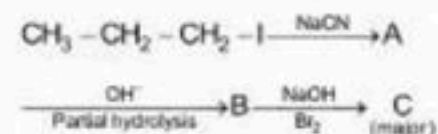
Question-51 Select the correct statement ?

Type	Multiple choice
Option	All canonical forms always contribute equally to the resonance hybrid. incorrect
Option	In both ethanamine and ethenamine nitrogen is sp^2 hybridised. incorrect
Option	All 'C-O' bond length in carbonate dianion are equal. correct
Option	$\text{CH}_2=\text{C}=\text{O}$ does not exhibit resonance because it is not a conjugated system. incorrect
Solution	 Equivalent resonating structures thus all bonds have equal bond lengths.

NEET 2024.

Test Series

96. Identify the major product C formed in the following reaction sequence:



- (1) propylamine (2) butylamine
(3) butanamide (4) α -bromobutanoic acid

Answer (1)

Question-46 In the Hoffmann bromamide degradation reaction

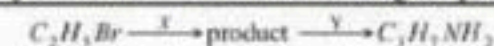


Intermediates are :

- a. RCONHBr
b. RNCN
c. RNH_2

Type	Multiple choice
Option	a only Incorrect
Option	b only Incorrect
Option	a and b both correct
Option	c only Incorrect
Solution	
Marks	4 1

Question-62 Identify X and Y in the following sequence



Type	Multiple choice
Option	$\text{X} = \text{KCN}, \text{Y} = \text{LiAlH}_4$ correct
Option	$\text{X} = \text{KCN}, \text{Y} = \text{H}_3\text{O}^+$ incorrect
Option	$\text{X} = \text{CH}_3\text{Cl}, \text{Y} = \text{AlCl}_3 / \text{HCl}$ incorrect
Option	$\text{X} = \text{CH}_3\text{NH}_2, \text{Y} = \text{HNO}_2$ incorrect
Solution	$\text{C}_2\text{H}_5\text{Br} \xrightarrow{\text{KCN/Y}} \text{C}_2\text{H}_5\text{CN} \xrightarrow{\text{LiAlH}_4/\text{H}^+} \text{C}_2\text{H}_5\text{CH}_2\text{NH}_2 (\text{C}_3\text{H}_7\text{NH}_2)$ $\text{X} = \text{KCN}, \text{Y} = \text{LiAlH}_4$



Same concept



NEET 2024.

Test Series

86. The pair of lanthanoid ions which are diamagnetic is

- (1) Ce^{4+} and Yb^{2+}
- (2) Ce^{3+} and Eu^{2+}
- (3) Gd^{3+} and Eu^{3+}
- (4) Pm^{3+} and Sm^{3+}

Answer (1)

Question-80 Which of the following lanthanoid ions are diamagnetic ? (Ce = 58, Sm = 62, Eu = 63, Yb = 70)

Type Multiple_choice

Option Yb^{+2} incorrect

Option Sm^{+2} correct

Option Ce^{+2} incorrect

Option Eu^{+2} incorrect

Solution

Marks 4

1



NEET 2024.

Test Series

65 Match List I with List II.

List I (Complex)

- A. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$
- B. $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$
- C. $[\text{Co}(\text{NH}_3)_5][\text{Cr}(\text{CN})_6]$
- D. $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$

List II (Type of isomerism)

- I. Solvate isomerism
- II. Linkage isomerism
- III. Ionization isomerism
- IV. Coordination isomerism

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-III, C-IV, D-II
- (3) A-I, B-IV, C-III, D-II
- (4) A-II, B-IV, C-III, D-I

Answer (1)

Question-87 The complex $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $[\text{Co}(\text{CN})_6][\text{Cr}(\text{NH}_3)_6]$ are example of

Type Multiple_choice

Option Ionization isomerism Incorrect

Option Coordination Isomerism correct

Option Geometrical Isomerism Incorrect

Option Linkage isomers Incorrect

Solution

Marks 4 1

NEET 2024.

Test Series

'Spin only' magnetic moment is same for which of the following ions?

- A. Ti^{3+} B. Cr^{2+}
 C. Mn^{2+} D. Fe^{2+}
 E. Sc^{3+}

Choose the most appropriate answer from the options given below.

- (1) B and D only
 (2) A and E only
 (3) B and C only
 (4) A and D only

Answer (1)

Question-60 The magnetic moment is associated with its spin angular momentum and orbital angular momentum. Spin only magnetic moment value of Cr^{3+} ion is :

Type	Multiple_choice	
Option	2.87	incorrect
Option	3.54	incorrect
Option	3.87	correct
Option	2.54	incorrect

Solution

Cr^{+3} has an electronic configuration of $[\text{Ar}] 3d^3 4s^0$, having unpaired electrons as 3. Hence magnetic moment of Cr^{+3} is $\sqrt{3(3+2)} = 3.87\text{BM}$

Marks 4 1



Same

concept



NEET 2024.

70. Which reaction is **NOT** a redox reaction?

- (1) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
- (2) $2\text{KClO}_3 + \text{I}_2 \rightarrow 2\text{KIO}_3 + \text{Cl}_2$
- (3) $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$
- (4) $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$

Answer (4)

Test Series

Question-56 Which of the following is not a Redox reaction

Type	Multiple_choice	
Option	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$	correct
Option	$\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + 1/2\text{H}_2$	inCorrect
Option	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$	Incorrect
Option	$\text{MnCl}_3 \rightarrow \text{MnCl}_2 + 1/2\text{Cl}_2$	Incorrect
Solution	No change in oxidation no so it is a redox reaction	
Marks	4	1



Same

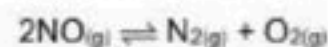
concept



NEET 2024.

Test Series

99. Consider the following reaction in a sealed vessel at equilibrium with concentrations of $N_2 = 3.0 \times 10^{-3} \text{ M}$, $O_2 = 4.2 \times 10^{-3} \text{ M}$ and $NO = 2.8 \times 10^{-3} \text{ M}$.



If 0.1 mol L^{-1} of $NO_{(g)}$ is taken in a closed vessel, what will be degree of dissociation (α) of $NO_{(g)}$ at equilibrium?

- (1) 0.00889 (2) 0.0889
(3) 0.8889 (4) 0.717

Answer (4)

Question-52 Calculate the degree of dissociation of HI as $2HI \rightleftharpoons H_2 + I_2$ at 450°C if the equilibrium constant for the dissociation reaction is 0.263

Type Multiple_choice

Option 0.688 InCorrect

Option 0.326 incorrect

Option 0.454 Incorrect

Option 0.253 correct

Solution

Marks 4

1



Same concept



NEET 2024.

Test Series

64. Match List I with List II.

List I

(Conversion)

- A. 1 mol of H_2O to O_2
- B. 1 mol of MnO_4^- to Mn^{2+}
- C. 1.5 mol of Ca from molten CaCl_2
- D. 1 mol of FeO to Fe_2O_3

List II

(Number of Faraday required)

- I. 3F
- II. 2F
- III. 1F
- IV. 5F

Choose the correct answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-III, B-IV, C-I, D-II
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-IV, C-II, D-I

Answer (1)

Question-51 In acidic medium MnO_4^- is converted to Mn^{2+} when acts as an oxidising agent. The quantity of electricity required to reduce 0.05 mole of MnO_4^- would be -

Type Multiple_choice

- | | | |
|--------|--------|-----------|
| Option | 0.01 F | incorrect |
| Option | 0.05 F | incorrect |
| Option | 0.25 F | correct |
| Option | 0.15 F | incorrect |

Solution

Marks 4

1



Same concept



NEET 2024.

Test Series

66. Match List I with List II.

List I

(Compound)

- A. NH_3
- B. BrF_5
- C. XeF_4
- D. SF_6

List II

(Shape/geometry)

- I. Trigonal Pyramidal
- II. Square Planar
- III. Octahedral
- IV. Square Pyramidal

Choose the correct answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
- (2) A-II, B-IV, C-III, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-IV, D-I

Answer (1)

Practice more than 50
questions in test series

One of the easiest question

There are more questions from test series but in one question of NEET 4-5 question will be require

Only 3-4 questions are not in tests but will try next year 50/50

Thank you for your love to
Rakshita Singh SBTS